



How can biochar-based metal oxide nanocomposites counter salt toxicity in plants?

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Abstract Application of biochar-based metal oxide nanocomposites can acquire new composites and combine the benefits of biochar with nanomaterials. For the first time, this research was conducted to evaluate the possible effects of solid biochar (25 g biochar kg⁻¹ soil) and biochar-based nanocomposites (BNCs) of magnesium oxide (25 g BNC-MgO kg⁻¹ soil), manganese oxide (25 g BNC-MnO biochar kg⁻¹ soil) and combined use of these nanocomposites (12.5 g BNC-MgO + 12.5 g BNC-MnO kg⁻¹ soil) on salt (non-saline, 6 and 12 dSm⁻¹ NaCl salinities) tolerance of safflower plants (*Carthamus tinctorius* L.). Salinity reduced potassium, magnesium and manganese contents in root and leaf tissues, chlorophyll content index, photosynthetic pigments, maximum quantum yield of photosystem II (Fv/Fm) and relative photosynthetic electron transport rate

(RETR), leaf water content and plant biomass, but increased the sodium content, reactive oxygen species generation (ROS), oxidative stress and antioxidants and ROS detoxification potential of safflower roots and leaves. Application of biochar and BNCs increased the contents of potassium, manganese and magnesium in plant tissues, photosynthetic pigments, Fv/Fm and RETR, leaf water content and reduced sodium accumulation, ROS generation and oxidative stress under saline conditions, leading to a higher plant biomass in comparison with control. The BNC-MgO + BNC-MnO was the superior treatment on reducing salt toxicity. This treatment reduced oxidative stress by enhancing photosynthetic pigments, Fv/Fm and RETR of safflower under salt stress. These results revealed that BNCs have a great potential for improving salt tolerance of plants through increasing RETR and decreasing sodium accumulation and ROS generation.

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Introduction

Environmental stresses such as salt toxicity of soil limit crop production worldwide. Salt toxicity

decreases crop growth and development through ion toxicities and imbalances, osmotic and oxidative stresses (Farhangi-Abriz and Ghassemi-Golezani 2018; Rady et al. 2019). As a consequence, stomatal conductivity, gas exchange capacity, photosynthetic activities and electron transport in cellular organelles decline, leading to a high generation of reactive oxygen species (ROS) such as superoxide radical $O_2^{\bullet-}$, singlet oxygen (1O_2), hydroxyl radical ($OH^\bullet$) and hydrogen peroxide (H_2O_2) (Liang et al. 2018). The ROS cause great harms to lipid membranes of plant cells and other vital macromolecules including nucleic acids and proteins and eventually induce oxidative stress and cell death (Bose et al. 2014; Farhangi-Abriz et al., 2020).

Stress-induced ROS generation is deactivated by enzymatic antioxidants such as catalase (CAT), superoxide dismutase (SOD), peroxidase (POX) and non-enzymatic compounds including tocopherols, ascorbic acid and phenolic compounds (Ali et al. 2017; Cheng et al. 2018). Among the various ROS, hydroxyl radical and singlet oxygen cannot hunt by enzymatic antioxidants, and detoxification of these compounds is exclusively reliant on non-enzymatic low molecular weight antioxidants such as ascorbic acid and tocopherols (Noctor et al. 2018). The adequate amount of nutrients is necessary for enhancing salt tolerance in plants. Nutrients are the mediators of metabolism and adjust numerous physiological pathways such as antioxidative activities under normal and stressful conditions. Several strategies such as enhancing soil nutrients by biochar have been applied to reduce the toxic effects of salt and/or other environmental stresses in plants (Ali et al. 2017; Rajput et al., 2020).

Biomass-derived charcoal or biochar is a carbon-rich material manufactured by thermal breakdown of biomass under low or anoxia conditions (Ghassemi-Golezani and Farhangi-Abriz 2019; Rehman et al. 2019). There is excellent potential for char materials such as biochar to close an agronomic circle through decreased energy inputs, improved yields, minimized carbon emissions to the atmosphere and reduced irrigation requirements. Current knowledge has confirmed that biochar addition to the farm lands can increase soil fertility, along with char material's chief role in climate change alleviation by keeping carbon in the pedosphere in a constant form (Ali et al. 2017; Zhang et al. 2019). Biochar is well known to increase

the crop tolerance against salt stress by improving soil properties (Amini et al. 2016), increasing nutrient availability, reducing sodium and enhancing potassium uptakes, and also preserving water potential in an optimum level for plant growth (Ali et al. 2017). Nutrient availability in biochar structure is completely dependent to the raw material types and pyrolysis condition. Biochar is a nutrient-rich material, but some of the key nutrients such as magnesium and manganese are deficient in this organic material (Novak et al. 2009; Naeem et al. 2014).

Magnesium is an important nutrient for crop growth and productivity. This nutrient as a part of chlorophyll has a role in activation of enzymes and preservation of ribosome molecular structure (Fageria 2016). The adequate amount of magnesium is essential for salt tolerance of plants (Chen et al. 2017). Manganese is another important nutrient in plant life cycle. This nutrient has numerous roles such as indirect effect on chlorophyll biosynthesis, increasing respiratory enzymes activities, stimulating calcium and phosphate uptake, enhancing germination and seedling growth, involving in the electron transport systems and oxidation–reduction reactions, nitrate assimilation and iron metabolism in plant cells (Fageria 2016). Manganese availability can increase salt tolerance in different plant species (Pandya et al. 2005). While nutrients are very vital for plant growth and productivity, most of the nutrients are rendered unobtainable to crops due to several reasons such as decomposition, leaching and hydrolysis (Fageria 2016). So, it is crucial to reduce nutrient losses in fertilization process and to enhance the crop productivity through the utilization of modern technologies with the help of nanoscience.

Nanomaterials, because of their special physicochemical properties, mainly high specific surface area, commonly have a very high fertilization capacity in agriculture and can provide adequate level of nutrients for plant growth (Servin et al. 2017). New technologies combining engineered nanomaterials into biochar-making processes could develop the functions of biochar for wide range applications including carbon sequestration, wastewater treatment and soil fertility enhancement (Tan et al. 2016; Zama et al. 2018). The composition process changes the physicochemical and molecular characters of biochar to create unique and more practical materials with well-defined properties (Zama et al. 2018). Producing biochar-based

nanocomposites of metal oxides can create new complex materials and combine the benefits of biochar with nanomaterials. The resulting composites frequently display great improvement in physicochemical properties such as pore numbers, functional groups, catalytic degradation ability, surface active areas, easy to separation and cation exchange capacity (Liu et al. 2015; Tan et al. 2016). Previous studies showed that enriching biochar with nutrients can enhance the plant growth and biochar performance. For example, nitrogen has been incorporated on biochar to produce biochar-based slow release nitrogen fertilizers with excellent water retention capacities (Wen et al. 2017). Enriching biochar with silicon mitigated the arsenic toxicity and improved the growth of spinach plants (Zama et al. 2018). Application of biochar-based nanocomposites might be a practical way for improving abiotic stress tolerance of plants. So, the objective of this novel research was to assess the possible effects of biochar-based nanocomposites of magnesium and manganese oxides on reducing oxidative stresses of salt toxicity in safflower (*Carthamus tinctorius* L.) plants.

Materials and methods

Production of biochar-based nanocomposites

Biochar-based nanocomposites of magnesium and manganese oxides were made by the methods of Shen et al. (2019) and Song et al. (2014), respectively. Briefly, a sample of biochar was mixed separately with 0.01 M magnesium chloride hexahydrate ($\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) and 0.01 M potassium permanganate (KMnO_4) at solid to liquid ratio of 1:40 g mL^{-1} and stirred for 4 h. The mixtures were oven-dried at 80 °C for 5 h and thereafter were heated at 460 °C. The sample was then washed thoroughly with distilled water to remove impurities and dried at 80 °C for 3 h. All of the chemicals were purchased from Sigma-Aldrich Chemicals company (Missouri, USA). The main properties of biochar and biochar-based nanocomposites of magnesium and manganese oxides are presented in Table 1.

The pH of biochar and biochar-based metal oxide nanocomposites in a 1:2.5 (w/v) ratio with deionized water was measured by a pH meter (Model: HI 99,121, Hanna Instrument, USA). The CHNOS elemental

analyzer (Elementar group, Hanau, Germany) was used to determine C, N, H and O contents of biochar. The cation exchange capacity was determined by the ammonium acetate method (Hendershot et al. 1993). The contents of nutrients were measured by ICP Optical Emission Spectrometry (5800 ICP-OES Instrument, Hitachi, Tokyo, Japan) (Zhang et al. 2017). The Quantachrome Autosorb Automated Gas Sorption System was used to determine the Brunauer–Emmett–Teller (BET) surface area and pore size distributions of char materials (White 2010). The particle sizes of MgO and MnO within the biochar matrix were calculated by the Scherrer formula (Yao et al. 2013). The TGA/DSC analysis was applied to estimate the thermal stability of biochar and metal oxide nanocomposites and maximum weight loss due to heat treatment (because of the CO_2 , CO and CH_4 emissions and decomposition of organic compounds). This was considered as maximum stability of these materials (Lateef et al. 2019). For assaying sodium sorption capacity, about 200 mg from each char material was added to 25 mL of 0.2 M NaCl solution, and then it was shaken for 12 h. The solution was filtered using Whatman 42 filter paper and the sodium concentration in the filtrated solution was measured by a Corning 410 flame photometer (Corning, New York, USA) (Rostamian et al. 2015). In this way, sodium sorption capacity of biochar and biochar-based metal oxide nanocomposites was determined.

Experimental conditions

This research was conducted in a greenhouse with the day and night mean temperatures of 24 and 20 °C, respectively, 140 Wm^{-2} light intensity, 39% air humidity and about 13-h photoperiod. The soil samples were collected from the upper layer (0–30 cm) of an orchard and analyzed according to Reynolds et al. (2008). The main properties of experimental soil are shown in Table 1. Three levels of sodium chloride salinities [non-saline (1.32 dSm^{-1}), 6 and 12 dSm^{-1} salinities were adjusted based on saturated soil paste extract] and five levels of biochar-based nanocomposites (non-biochar, 25 g kg^{-1} biochar, 25 g kg^{-1} biochar-based nanocomposite of magnesium oxide, 25 g kg^{-1} biochar-based nanocomposite of manganese oxide and 12.5 g kg^{-1} biochar-based nanocomposite of magnesium oxide + 12.5 g kg^{-1} biochar-based

Table 1 Some physico-chemical properties of the experimental soil, biochar and biochar-based nanocomposites of magnesium (BNC-MgO) and manganese oxides (BNC-MnO)

Soil		Biochar		BNC-MgO		BNC-MnO	
Texture	Silty loam	N (%)	0.52	N (%)	0.58	N (%)	0.54
pH	6.9	C (%)	44.1	C (%)	40.25	C (%)	39.33
EC (dSm ⁻¹)	1.32	H (%)	2.7	H (%)	1.88	H (%)	2.35
OC (g kg ⁻¹)	11.1	O (%)	27.40	O (%)	28.95	O (%)	29.40
Total N (%)	0.09	P (mg kg ⁻¹)	8250	P (mg kg ⁻¹)	8120	P (mg kg ⁻¹)	7320
P (mg kg ⁻¹)	31.2	S (mg kg ⁻¹)	5790	S (mg kg ⁻¹)	5560	S (mg kg ⁻¹)	5340
K (mg kg ⁻¹)	187.1	Na (mg kg ⁻¹)	2.30	Na (mg kg ⁻¹)	1.80	Na (mg kg ⁻¹)	1.82
Mg (mg kg ⁻¹)	70.4	K (mg kg ⁻¹)	4070	K (mg kg ⁻¹)	3610	K (mg kg ⁻¹)	3990
Mn (mg kg ⁻¹)	10.80	Ca (mg kg ⁻¹)	3750	Ca (mg kg ⁻¹)	3640	Ca (mg kg ⁻¹)	3240
CEC (cmol kg ⁻¹)	18.80	Mg (mg kg ⁻¹)	166.4	Mg (mg kg ⁻¹)	3430	Mg (mg kg ⁻¹)	112.3
Moisture (%)	1	Mn (mg kg ⁻¹)	37.30	Mn (mg kg ⁻¹)	29.64	Mn (mg kg ⁻¹)	2250
		Cl (mg kg ⁻¹)	N. A	Cl (mg kg ⁻¹)	10.40	Cl (mg kg ⁻¹)	N. A
		CEC (cmol kg ⁻¹)	20.21	CEC (cmol kg ⁻¹)	25.20	CEC (cmol kg ⁻¹)	23.60
		pH	6.8	pH	7.1	pH	7.6
		S _{BET} (m ² g ⁻¹)	14.2	S _{BET} (m ² g ⁻¹)	108.2	S _{BET} (m ² g ⁻¹)	65.4
		V _{tot} (cm ³ g ⁻¹)	0.024	V _{tot} (cm ³ g ⁻¹)	0.106	V _{tot} (cm ³ g ⁻¹)	0.088
		AVP (nm)	3.43	AVP (nm)	3.81	AVP (nm)	5.57
		Stability (°C)	510	MgO size (nm)	23–60	MnO size (nm)	35–68
		Na-SC (mg g ⁻¹)	46.7	Stability (°C)	490	Stability (°C)	490
		Moisture (%)	< 1	Na-SC (mg g ⁻¹)	91.4	Na-SC (mg g ⁻¹)	73.6
				Moisture (%)	< 1	Moisture (%)	< 1

EC Electrical conductivity; OC organic carbon; CEC cation exchange capacity; NA not available; S_{BET} BET surface area; V_{tot} total pore volume; AVP average pore size; Na-SC sodium sorption capacity

nanocomposite of manganese oxide) were applied. The soil was well mixed with biochar and nanocomposites according to the treatments and transferred to the polyethylene pots (25 × 25 cm). Subsequently, about 10 seeds of *Carthamus tinctorius* L. cv. Sina were sown in each pot and irrigated with tap water to achieve 100% field capacity (FC). For plant nutrition, 5 g of a fertilizer (Master Valagro, Italy, with N: 20%, P₂O₅: 20%, K₂O: 20%, B: 0.02%, Mn: 0.03%, Zn: 0.01%, Cu: 0.005% and Fe: 0.07%) was dissolved in a liter of tap water (EC = 0.6 dSm⁻¹ and pH = 7.1) and added to the pots (250 mL per pot). After seedling emergence, the plants were thinned and only four plants were kept in each pot. At flowering stage, two plants from each pot were carefully taken out and washed with distilled water (root and shoot) and morphological and physiological measurements were taken. For physiological measurements, plant samples

were frizzed with liquid nitrogen and kept under −80 °C for future measurements.

Nutrient content

The dry ash method was used to assess the sodium, potassium, magnesium and manganese contents of safflower plants (Jones and Case 1990). Safflower roots and leaves were oven-dried at 80 °C and powdered. Then, 200 mg from each sample was burned at 560 °C for 5 h and the ashes were digested in 10 mL 1 M HCl at 25 °C for 24 h. Nutrients were measured by atomic absorption spectrophotometry (Shimadzu model: AA-7000, Kyoto, Japan).

Leaf characters and plant biomass

Photochemical efficiency of photosystem II (Fv/Fm) and relative photosynthetic electron transport rate

(RETR) was measured using a portable chlorophyll fluorometer (OS-30, Opti-Sciences, USA). Safflower leaves were separated and weighed (FW) and then oven-dried at 80 °C for 48 h and reweighed (DW). The leaf water content (LWC) was determined by the following formula:

$$\text{LWC} = [(\text{FW} - \text{DW}) \div \text{FW}] \times 100$$

Leaf CCI (chlorophyll content index) was assayed using a portable chlorophyll meter (CCM-200, Opti-Sciences, USA). Anthocyanin content was determined by the method of Mancinelli (1984). Fresh leaf or root samples (500 mg each) were separately homogenized in 1.5 ml extraction buffer containing 0.3 ml water, 1.1 ml methanol and 0.01 ml HCl. Homogenated samples were centrifuged at 12,000 g for 20 min at 4 °C. The absorbance of supernatant was read at 530 and 657 nm (Dynamica, Halo DB-20-UV-Visible Spectrophotometer, UK). Flavonoid and carotenoid contents were determined by the method of Zhishen et al. (1999) and Maclachlan and Zalik (1963), respectively. Safflower plant biomass was determined after oven-drying at 80 °C for 48 h.

Antioxidant enzymes

Roots and young leaves of safflower were harvested, and enzyme activities were assayed. The enzymes were extracted from 250 mg of roots or leaves by potassium phosphate buffer with adjusted pH (pH = 7). Extracted samples were centrifuged at 15,000 g at 4 °C for 15 min, and the supernatant was collected and used for the enzyme assay. The POX activity in safflower leaves and roots was determined by an increase in absorption at 470 nm due to guaiacol oxidation for 2 min. The reaction buffer contained 1.2 ml of 50 mM potassium phosphate buffer (pH 7.0), 0.5 ml of 1% H₂O₂, 0.5 ml of 1% guaiacol and 0.15 ml of enzyme extract (Gueta-Dahan et al. 1997). The CAT activity was assayed by the method of Farhangi-Abriz and Ghassemi-Golezani (2018) with monitoring the changes in absorbance at 240 nm. The reaction buffer for this assay contained 50 mmol L⁻¹ phosphate buffer (pH 7.0), 30 mmol L⁻¹ H₂O₂ and 0.1 mL of extracted enzyme. The SOD activity in plant tissues was estimated according to Dhindsa et al. (1981). One unit of this enzyme activity was assayed as the volume of enzyme causing 50% of the maximum inhibition of nitro blue tetrazolium. Soluble

protein contents in safflower leaves and roots were assayed by the method of Bradford (1976), and enzymes activities were expressed as U g⁻¹ FW.

Phenols, ascorbic acid and α -tocopherol

Total of phenolic compounds in safflower leaf and root was determined by the Folin–Ciocalteu reagent according to the method of Ainsworth and Gillespie (2007). Initially, 0.1 g of root or leaf samples was homogenized in 5 ml methanol 80%, and then 2.5 ml of methanolic-extracted samples was mixed with 1.25 ml Folin–Ciocalteu reagent. After 2 h of incubation of the samples at room temperature (25 °C), the absorbance was read at 765 nm. The total phenol content in safflower root and leaf was determined by the standard curve of gallic acid. Ascorbic acid content was assayed by the method of Keller and Schwager (1977). About 0.1 g of leaf or root sample was homogenized in 2 ml of oxalic acid and Na-EDTA extraction buffer. The homogenates were centrifuged at 20,000 g for 15 min, and 1 ml of supernatant was collected and 5 ml of 20 µg ml⁻¹ of 2,6-dichlorophenol-indophenol was added to each sample, and the absorbance was recorded at 520 nm. The α -tocopherol content was determined according to the method of Souid et al. (2018) and was expressed in µg g⁻¹ fresh weight.

Lipid peroxidation, ROS generation and ROS detoxification potential

Lipid peroxidation was determined according to the methods of Hemeda and Klein (1990) and described as thiobarbituric acid (TBA)-reactive substances (TBARS) content in plant tissues. In total, 100 mg of fresh leaf or root samples was homogenated in 2.5 mL of 0.1% (w/v) trichloroacetic acid (TCA) at 4 °C. The homogenated samples were centrifuged at 12,000 g for 10 min, and 0.5 ml supernatant was collected and added to 2 mL of 0.5% (w/v) thiobarbituric acid. The resultant samples were heated in a Bain Marie for 10 min at 100 °C, and after cooling at 4 °C, the absorbance was recorded at 532 nm and was corrected for non-specific turbidity by subtraction of the absorbance at 600 nm. H₂O₂ content in roots and leaves of safflower was determined by the method of Hsu and Kao (2007). About 100 mg of plant tissues was homogenized in 2 mL of 0.1% trichloroacetic

acid at 4 °C and centrifuged at 12,000 g for 10 min. In total, 0.5 mL of the supernatant was collected and added to the 1 mL of potassium iodide (1 M) plus 0.5 mL of potassium phosphate buffer (pH 7) and the absorbance was read at 390 nm. The $O_2^{\bullet-}$ generation in root or leaf tissues was determined according to Wang and Jiao (2000). Oxygen radical absorbance capacity (ORAC) and 2,2-diphenyl-1-picrylhydrazyl (DPPH) were assayed according to the methods of Bacchiocca et al. (2006) and Hatano et al. (1988), respectively. The results were presented as ROS detoxification potential of plants.

Statistical analysis

The data were analyzed according to the experimental design (two-way ANOVA—factorial arrangement on the basis of randomized complete block design with three replicates) by the MSTAT-C software (East Lansing, Michigan State University, USA), and means were compared by Duncan's multiple range test at $p \leq 0.05$. Principal component analyses (PCA Pearson's correlation) for safflower growth, nutrients and antioxidants of roots and leaves were performed, using the XLSTAT 2017 software (Addinsoft, New York, USA). This multivariate analysis reveals the association of different parameters with each other. The position of each parameter on PCA coordinates indicates the relative impact of this parameter on other parameters. All figures were drawn by the Excel 2019 (Microsoft Corporation, USA).

Results

Nutrient content

Enhancing salt stress increased sodium, but reduced potassium and magnesium contents in safflower roots and leaves (Table 2). Increasing salt toxicity of soil up to 6 dSm⁻¹ had no significant effect on root and leaf manganese content in safflower plants, but it was decreased by further increment of salt toxicity up to 12 dSm⁻¹. Application of solid biochar and biochar-based metal oxide nanocomposites did not change sodium content of roots and leaves of safflower under non-saline condition, but these treatments noticeably reduced sodium contents under saline conditions. Biochar-based metal oxide nanocomposites had a

better effect on reducing sodium content in root and leaf tissues, compared with solid biochar.

Soil treatment with solid biochar and biochar-based metal oxide nanocomposites significantly improved the potassium, magnesium and manganese contents in safflower roots and leaves (Table 2). These improvements were occurred in both saline and non-saline conditions. In most cases, the highest potassium content was achieved by biochar-based nanocomposites of magnesium oxide and combined use of nanocomposites under saline and non-saline conditions. Application of biochar-based metal oxide nanocomposites in combined form improved the potassium content of roots and leaves under severe stress by about 37% in comparison with the non-biochar treatments.

The highest magnesium content in root and leaf tissues was achieved by biochar-based nanocomposite of magnesium oxide under all levels of salinity. In most cases, combined use of metal oxide nanocomposites was ranked as the second superior treatment on rising magnesium content in plant tissues, followed by solid biochar and manganese nanocomposite. The highest manganese content of root and leaf tissues was recorded for biochar-based nanocomposite of manganese oxide and combined form of metal oxide nanocomposites. The biochar-based nanocomposite of magnesium had no significant impact on manganese content of safflower plants, compared to the solid biochar treatment. The combined form of metal oxide nanocomposites enriched the safflower roots and leaves with manganese by about 123% and 72% under high level of salinity and by about 101% and 91% under non-saline condition, respectively (Table 2).

Leaf water content (LWC) and plant biomass

Increasing salinity reduced LWC of safflower (Table 3). Application of biochar and biochar-based nanocomposites increased LWC of safflower under different levels of salt toxicity, but had no significant effect on this parameter in non-saline condition. Plant biomass was reduced by increasing salt stress. Application of biochar and biochar-based nanocomposites increased plant biomass under non-saline and saline conditions. Combined use of biochar-based nanocomposites of magnesium and manganese oxides had better effect than other treatments on improving plant biomass. This improvement was about 29% and 15%

Table 2 Means of some nutrients content (mg g^{-1} DW) in safflower plants affected by different levels of salt stress and biochar-based nanocomposites of magnesium and manganese oxides

Salinity	Soil treatments	Sodium		Potassium		Magnesium		Manganese	
		Root	Leaf	Root	Leaf	Root	Leaf	Root	Leaf
0 (dSm^{-1})	Non-biochar	7.31 $\pm$ 0.88 g	5.29 $\pm$ 0.89 e	33.37 $\pm$ 0.92 c	35.68 $\pm$ 1.21 b	6.00 $\pm$ 0.57 fgh	6.65 $\pm$ 0.34 efg	2.31 $\pm$ 0.29 c	1.34 $\pm$ 0.36 cd
	Biochar	8.28 $\pm$ 0.91 g	5.58 $\pm$ 0.77 e	38.00 $\pm$ 1.73 b	38.61 $\pm$ 1.25 a	8.00 $\pm$ 0.55 bcd	8.35 $\pm$ 0.35 bc	3.67 $\pm$ 0.36 b	2.35 $\pm$ 0.29 b
	BNC-MgO	7.40 $\pm$ 0.84 g	5.31 $\pm$ 0.82 e	41.65 $\pm$ 1.20 a	41.58 $\pm$ 1.36 a	9.67 $\pm$ 0.33 a	9.31 $\pm$ 0.28 a	3.66 $\pm$ 0.31 b	2.31 $\pm$ 0.30 b
	BNC-MnO	8.36 $\pm$ 1.05 g	5.69 $\pm$ 0.65 e	38.70 $\pm$ 1.20 b	40.71 $\pm$ 1.26 a	8.00 $\pm$ 0.54 bcd	8.67 $\pm$ 0.36 ab	4.59 $\pm$ 0.40 a	3.00 $\pm$ 0.58 a
	BNC-MgO + BNC-MnO	8.00 $\pm$ 1.15 g	5.65 $\pm$ 0.1.01e	40.30 $\pm$ 1.32 ab	41.02 $\pm$ 0.98 a	9.00 $\pm$ 0.56 ab	8.64 $\pm$ 0.32 ab	4.67 $\pm$ 0.37 a	2.37 $\pm$ 0.33 b
6 (dSm^{-1})	Non-biochar	26.68 $\pm$ 0.67 d	24.07 $\pm$ 1.00 c	25.02 $\pm$ 0.58 fg	25.64 $\pm$ 0.89 ef	5.34 $\pm$ 0.34 hi	6.01 $\pm$ 0.53 g	2.32 $\pm$ 0.32 c	1.17 $\pm$ 0.44 cd
	Biochar	22.67 $\pm$ 0.75 e	19.02 $\pm$ 1.00 d	29.00 $\pm$ 0.64 de	31.00 $\pm$ 0.58 cd	7.00 $\pm$ 0.55 def	7.34 $\pm$ 0.35 de	3.35 $\pm$ 0.31 b	2.10 $\pm$ 0.50 b
	BNC-MgO	20.70 $\pm$ 0.65 ef	16.61 $\pm$ 0.93 d	32.67 $\pm$ 0.88 c	34.05 $\pm$ 0.74 bc	8.67 $\pm$ 0.35 abc	9.32 $\pm$ 0.32 a	3.68 $\pm$ 0.21 b	2.13 $\pm$ 0.46 b
	BNC-MnO	19.61 $\pm$ 0.72 f	17.23 $\pm$ 0.78 d	31.62 $\pm$ 0.74 cd	31.32 $\pm$ 0.84 cd	6.67 $\pm$ 0.29 efg	7.00 $\pm$ 0.58 def	4.28 $\pm$ 0.36 a	3.03 $\pm$ 0.54 a
	BNC-MgO + BNC-MnO	20.31 $\pm$ 0.74 f	16.55 $\pm$ 0.44 d	33.00 $\pm$ 0.84 c	31.29 $\pm$ 0.99 cd	7.67 $\pm$ 0.34 cde	7.66 $\pm$ 0.67 cd	4.71 $\pm$ 0.32 a	2.93 $\pm$ 0.63 a
12 (dSm^{-1})	Non-biochar	37.35 $\pm$ 0.64 a	33.07 $\pm$ 0.59 a	20.06 $\pm$ 0.34 h	20.70 $\pm$ 0.63 g	3.00 $\pm$ 0.57 j	4.34 $\pm$ 0.33 h	1.70 $\pm$ 0.19 d	0.80 $\pm$ 0.35 d
	Biochar	31.29 $\pm$ 0.18 b	27.00 $\pm$ 0.55 b	24.00 $\pm$ 0.74 g	25.03 $\pm$ 0.82 f	5.00 $\pm$ 0.58 hi	6.00 $\pm$ 0.53 g	2.66 $\pm$ 0.28 c	1.20 $\pm$ 0.30 cd
	BNC-MgO	29.05 $\pm$ 0.43 c	24.36 $\pm$ 0.82 c	26.58 $\pm$ 0.59 efg	28.00 $\pm$ 1.04 ef	6.00 $\pm$ 0.54 fgh	7.31 $\pm$ 0.37 de	2.68 $\pm$ 0.27 c	1.20 $\pm$ 0.30 cd
	BNC-MnO	28.89 $\pm$ 0.69 c	24.23 $\pm$ 0.47 c	25.35 $\pm$ 0.93 fg	26.06 $\pm$ 0.64 ef	4.67 $\pm$ 0.35 i	6.34 $\pm$ 0.28 fg	3.35 $\pm$ 0.35 b	1.51 $\pm$ 0.28 c
	BNC-MgO + BNC-MnO	29.00 $\pm$ 0.56 c	22.41 $\pm$ 0.66 c	27.61 $\pm$ 0.58 ef	28.34 $\pm$ 0.86 de	5.67 $\pm$ 0.28 ghi	7.00 $\pm$ 0.52 def	3.72 $\pm$ 0.34 b	1.40 $\pm$ 0.30 c

Data represent the average of three replicates ($n = 3$) $\pm$ standard error.

BNC-MgO Biochar-based nanocomposite of magnesium oxide; *BNC-MnO* biochar-based nanocomposite of manganese oxide; *AR* absorption rate

Different letters indicate significant differences by Duncan's multiple range test at $p \leq 0.05$

Table 3 Plant biomass (g), leaf water content (%), efficiency of photosystem II (Fv/Fm), relative photosynthetic electron transport rate (RETR), chlorophyll content index (CCI) and the content of photosynthetic pigments (mg g^{-1} DW) in safflower plants for the interaction of salinity and biochar-based nanocomposites

Salinity (dS m^{-1})	Soil treatments	Plant biomass	Leaf water	Fv/Fm	RETR	CCI	Flavonoid	Carotenoid	Anthocyanin
0 (dS m^{-1})	Non-biochar	10.77 $\pm$ 0.34 d	83.06 $\pm$ 0.52 a	0.74 $\pm$ 0.01 e	57.50 $\pm$ 0.24 e	22.56 $\pm$ 0.45 bcd	0.51 $\pm$ 0.01 c	0.56 $\pm$ 0.02 e	0.09 $\pm$ 0.01 c
	Biochar	12.07 $\pm$ 0.32 c	83.10 $\pm$ 0.49 a	0.77 $\pm$ 0.01 bcd	61.36 $\pm$ 0.31 d	23.76 $\pm$ 0.28 ab	0.52 $\pm$ 0.01 bc	0.68 $\pm$ 0.01 b	0.12 $\pm$ 0.01 b
	BNC-MgO	13.23 $\pm$ 0.29 b	83.10 $\pm$ 0.47 a	0.78 $\pm$ 0.01 abcd	67.40 $\pm$ 0.19 a	24.53 $\pm$ 0.32 a	0.54 $\pm$ 0.01 ab	0.71 $\pm$ 0.02 a	0.12 $\pm$ 0.01 b
	BNC-MnO	12.81 $\pm$ 0.36 b	84.03 $\pm$ 0.55 a	0.78 $\pm$ 0.01 abc	62.43 $\pm$ 0.26 c	22.90 $\pm$ 0.61 bc	0.54 $\pm$ 0.01 ab	0.66 $\pm$ 0.01 b	0.17 $\pm$ 0.02 a
	BNC-MgO + BNC-MnO	14.21 $\pm$ 0.27 a	83.03 $\pm$ 0.54 a	0.79 $\pm$ 0.02 a	65.60 $\pm$ 0.23 b	24.20 $\pm$ 0.15 a	0.55 $\pm$ 0.01 a	0.73 $\pm$ 0.01 a	0.13 $\pm$ 0.01 b
6 (dS m^{-1})	Non-biochar	7.62 $\pm$ 0.20 hi	71.46 $\pm$ 0.27 c	0.70 $\pm$ 0.01 f	52.36 $\pm$ 0.44 g	20.36 $\pm$ 0.44 f	0.47 $\pm$ 0.01 defg	0.48 $\pm$ 0.02 g	0.06 $\pm$ 0.01 d
	Biochar	8.45 $\pm$ 0.28 fg	73.66 $\pm$ 0.16 b	0.77 $\pm$ 0.02 cd	55.46 $\pm$ 0.28 f	22.06 $\pm$ 0.37 cde	0.48 $\pm$ 0.01 def	0.52 $\pm$ 0.01 f	0.08 $\pm$ 0.01 c
	BNC-MgO	8.99 $\pm$ 0.07 f	74.33 $\pm$ 0.34 b	0.77 $\pm$ 0.01 cd	57.46 $\pm$ 0.28 e	23.33 $\pm$ 0.57 ab	0.49 $\pm$ 0.02 de	0.61 $\pm$ 0.02 d	0.08 $\pm$ 0.01 c
	BNC-MnO	9.03 $\pm$ 0.21 f	74.33 $\pm$ 0.33 b	0.76 $\pm$ 0.01 de	56.20 $\pm$ 0.41 f	22.66 $\pm$ 0.38 bcd	0.48 $\pm$ 0.01 de	0.57 $\pm$ 0.01 e	0.08 $\pm$ 0.01 c
	BNC-MgO + BNC-MnO	9.73 $\pm$ 0.26 e	74.46 $\pm$ 0.26 b	0.79 $\pm$ 0.02 ab	60.86 $\pm$ 0.38 d	23.36 $\pm$ 0.27 ab	0.49 $\pm$ 0.01 d	0.64 $\pm$ 0.01 c	0.08 $\pm$ 0.01 c
12 (dS m^{-1})	Non-biochar	6.36 $\pm$ 0.39 j	63.40 $\pm$ 0.70 f	0.60 $\pm$ 0.01 i	41.40 $\pm$ 0.31 k	18.60 $\pm$ 0.42 g	0.45 $\pm$ 0.01 h	0.34 $\pm$ 0.01 j	0.04 $\pm$ 0.01 e
	Biochar	7.23 $\pm$ 0.47 i	66.50 $\pm$ 0.77 e	0.63 $\pm$ 0.02 h	44.00 $\pm$ 0.35 j	20.33 $\pm$ 0.38 f	0.46 $\pm$ 0.01 gh	0.40 $\pm$ 0.01 i	0.05 $\pm$ 0.01 de
	BNC-MgO	7.58 $\pm$ 0.40 hi	67.36 $\pm$ 0.68 de	0.66 $\pm$ 0.02 g	47.66 $\pm$ 0.31 hi	21.60 $\pm$ 0.42 de	0.47 $\pm$ 0.01 efg	0.45 $\pm$ 0.02 h	0.06 $\pm$ 0.01 d
	BNC-MnO	7.52 $\pm$ 0.32 hi	66.96 $\pm$ 0.54 de	0.65 $\pm$ 0.01 g	47.03 $\pm$ 0.38 i	21.13 $\pm$ 0.38 ef	0.47 $\pm$ 0.01 fgh	0.43 $\pm$ 0.01 h	0.06 $\pm$ 0.01 d
	BNC-MgO + BNC-MnO	8.21 $\pm$ 0.45 gh	68.40 $\pm$ 0.69 d	0.67 $\pm$ 0.01 g	52.12 $\pm$ 0.41 g	22.66 $\pm$ 0.61 bcd	0.48 $\pm$ 0.01 defg	0.48 $\pm$ 0.02 g	0.06 $\pm$ 0.01 d

Data represent the average of three replicates ($n = 3$) $\pm$ standard error.*BNC-MgO* Biochar-based nanocomposite of magnesium oxide; *BNC-MnO* biochar-based nanocomposite of manganese oxide; *DW* dry weight
Different letters in each column indicate significant difference at $p \leq 0.05$.

(under 12 dSm^{-1}) compared with the non-biochar and biochar treatments, respectively.

Fv/Fm and RETR

Salt toxicity and biochar-based nanocomposites had significant effects on Fv/Fm and RETR in safflower leaves. Increasing salt toxicity up to 6 dSm^{-1} did not alter the Fv/Fm value in safflower leaves, but it was reduced under 12 dSm^{-1} salt toxicity (Table 3). RETR value of safflower leaves was reduced by rising salt stress. Biochar and biochar-based nanocomposites enhanced the Fv/Fm and RETR values of safflower leaves under different levels of salt stress and non-saline condition. In general, combined use of biochar-based nanocomposites and biochar-based nanocomposite of magnesium oxide showed better effect than other treatments in improving Fv/Fm and RETR values under saline and non-saline conditions. Under the high level of salt toxicity, combined treatment of biochar-based nanocomposite of magnesium and manganese oxides improved the RETR by about 25%, compared with the non-biochar treatment.

CCI and photosynthetic pigments

The interaction of salinity and biochar-based nanocomposites was significant for CCI, flavonoid, carotenoid and anthocyanin contents ($p \leq 0.01$). These parameters were significantly decreased by increasing salt toxicity (Table 3). Application of biochar and biochar-based nanocomposites of magnesium and manganese oxides increased the CCI and photosynthetic pigments under saline and non-saline conditions. In most cases, the highest CCI and photosynthetic pigments of safflower leaves were recorded for biochar-based nanocomposite of magnesium oxide and combined use of nanocomposites. Biochar and biochar-based nanocomposite treatments showed a similar effect on rising flavonoid and anthocyanin contents in safflower leaves. However, the highest flavonoid content in non-saline condition was observed in plants treated with the combination of nanocomposites. Biochar-based nanocomposite treatments (specially combined use of nanocomposites) exposed the better effect than biochar on rising carotenoid content in different saline and non-saline conditions.

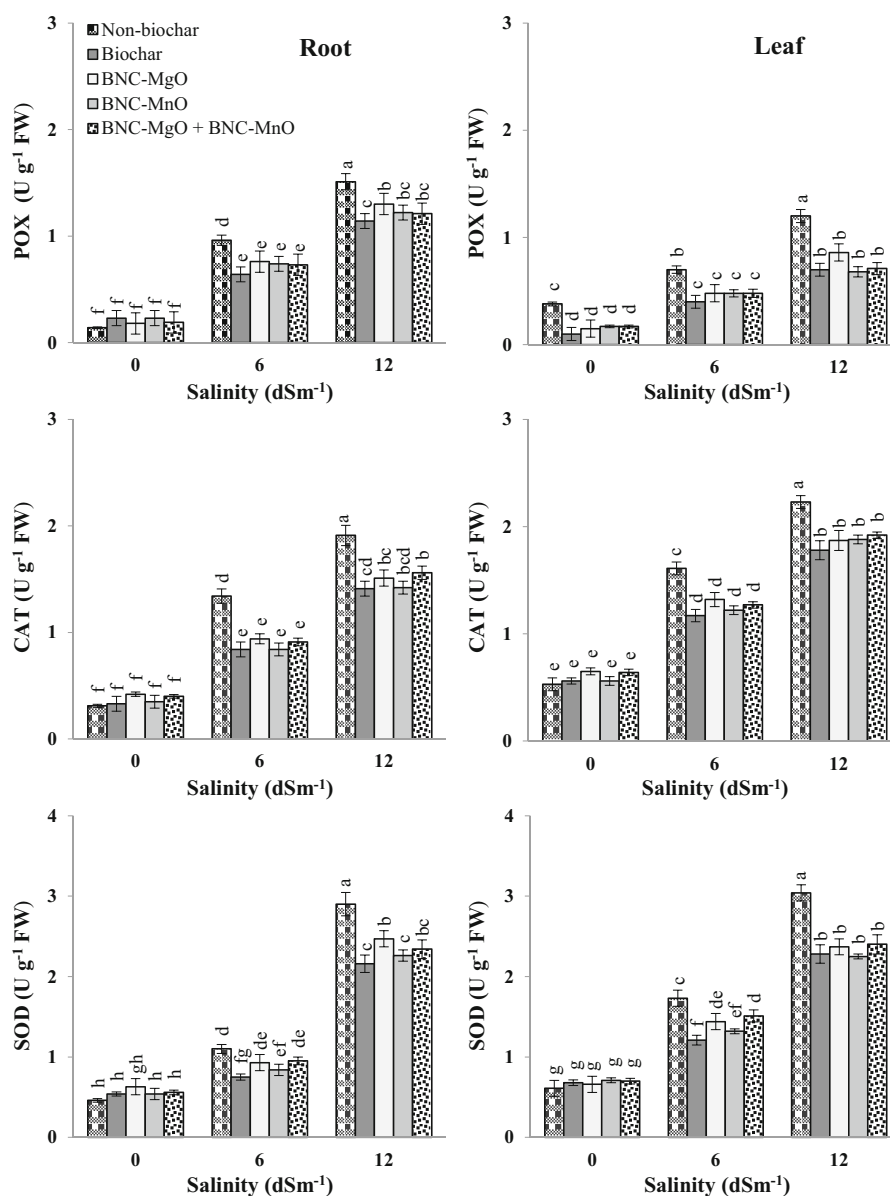
Antioxidant enzymes

The POX, CAT and SOD activities in safflower roots and leaves were enhanced, with increasing salt stress (Fig. 1). Application of biochar and biochar-based nanocomposite treatments reduced the POX, CAT and SOD activities in safflower roots and leaves under different levels of salt stress, but these treatments did not alter the antioxidant activities under non-saline condition. Meanwhile, POX activity in safflower leaves was reduced by biochar and nanocomposite treatments under non-saline condition. Nanocomposite treatments enhanced the POX, CAT and SOD activities in safflower roots and leaves, in comparison with the biochar treatments. However, these increments in most cases were not significant. In general, nanocomposite of magnesium oxide and combined use of nanocomposites (magnesium oxide + manganese oxide) had the highest effect on the activities of POX, CAT and SOD in roots and leaves of safflower. However, in some cases there was no significant difference between these treatments.

Ascorbic acid, phenolic compounds and α -tocopherol

There was a significant interaction between the salt stress and nanocomposites for ascorbic acid, phenolic compounds and α -tocopherol contents. These non-enzymatic antioxidants were increased by enhancing salt stress (Fig. 2). Biochar and biochar-based nanocomposites increased the ascorbic acid content of leaves and roots under non-saline condition; however, these treatments decreased the ascorbic acid content in both roots and leaves under saline conditions. In most cases, biochar-based nanocomposite of magnesium oxide and combined use of nanocomposites were resulted in the highest ascorbic acid content of roots and leaves. The phenolic compounds in safflower roots and leaves were reduced by the biochar and nanocomposite treatments under saline condition. Application of biochar and biochar-based nanocomposites under non-saline condition, did not alter the phenolic compounds in roots, but biochar-based nanocomposite of manganese oxide and combined use of nanocomposites significantly increased the phenolic compounds in leaves. Combined use of biochar-based nanocomposites of magnesium and manganese oxides was led to the highest content of

Fig. 1 Changes in POX, CAT and SOD activities in safflower roots and leaves in response to the biochar-based nanocomposites of magnesium and manganese under different levels of NaCl salinity. Data represent the average of three replicates ($n = 3$) $\pm$ standard error. Different letters indicate significant differences by Duncan multiple range test at $p \leq 0.05$. *POX* peroxidase; *CAT* catalase; *SOD* superoxide dismutase, *U* unit; *FW* fresh weight; *BNC-MgO* biochar-based nanocomposite of magnesium oxide; *BNC-MnO* biochar-based nanocomposite of manganese oxide

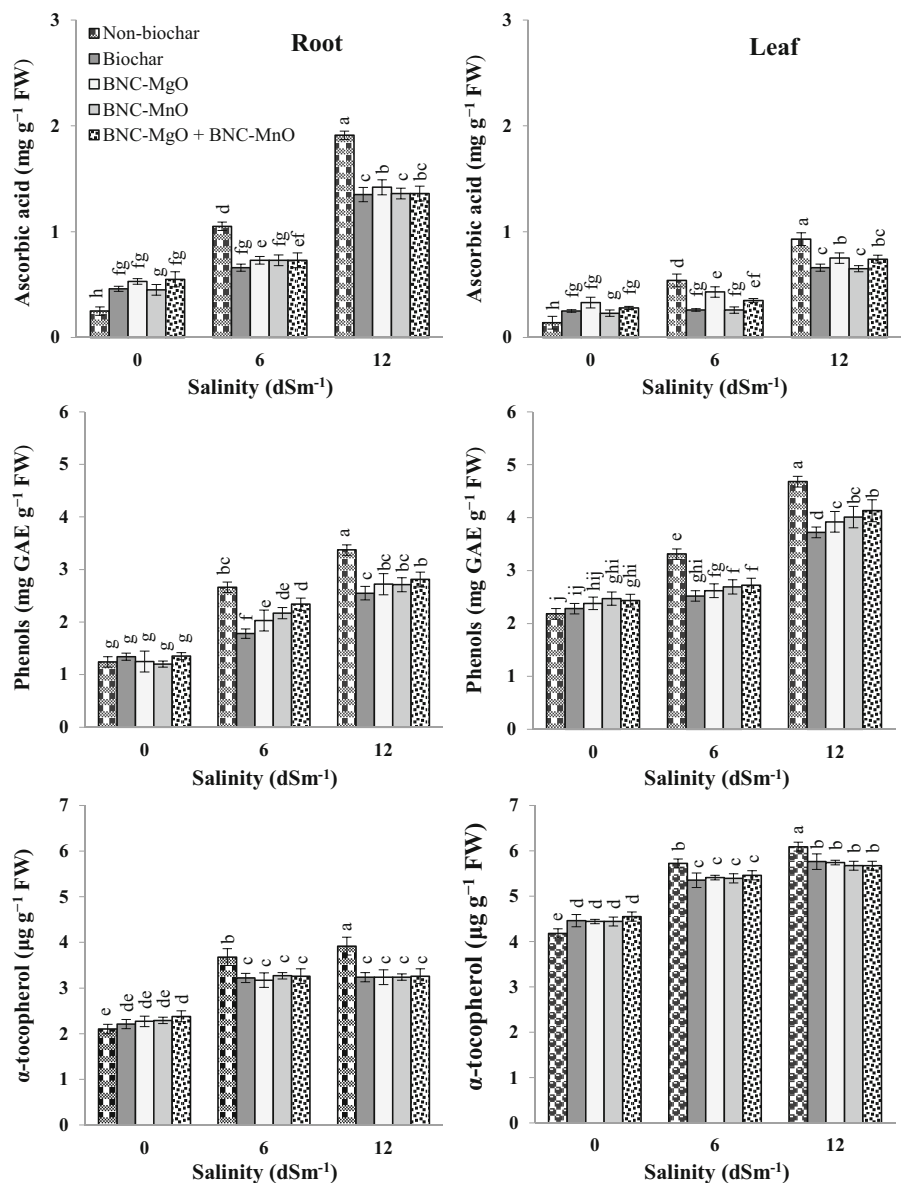


phenolic compounds, compared with different biochar and nanocomposite applications. Application of biochar and biochar-based nanocomposites of magnesium and manganese oxides improved α -tocopherol content of roots and leaves under non-saline condition. However, these treatments reduced this antioxidant under salt stress. All of biochar and biochar-based nanocomposite treatments showed a similar effect on decreasing α -tocopherol content in safflower leaves and roots.

Lipid peroxidation and ROS generation

Rising salinity intensified the TBARS content and ROS generation in leaves and roots of safflower (Table 4). Biochar and biochar-based nanocomposite treatments considerably reduced the TBARS content and ROS generation under salt stress. However, these treatments did not show a tangible effect on TBARS content and ROS generation under non-saline condition. Combined use of biochar-based nanocomposites of magnesium + manganese oxides caused the largest

Fig. 2 Means of ascorbic acid, phenolic compounds and α -tocopherol contents in safflower roots and leaves under different levels of salinity and biochar-based nanocomposites of magnesium and manganese. Data represent the average of three replicates ($n = 3$) $\pm$ standard error. Different letters indicate significant differences by Duncan multiple range test at $p \leq 0.05$. GAE Gallic acid equivalent; FW fresh weight; *BNC-MgO* biochar-based nanocomposite of magnesium oxide; *BNC-MnO* biochar-based nanocomposite of manganese oxide



reduction in TBARS and ROS generation in safflower plants. These superior effects on reducing ROS generation and lipid peroxidation were much obvious in high level of salt toxicity (12 dSm⁻¹). TBARS content under 12 dSm⁻¹ showed an impressive decrement in response to the combined use of biochar-based nanocomposites of magnesium + manganese oxides.

ROS detoxification potential

ORAC and DPPH were increased by rising salt stress (Table 4). Application of biochar and biochar-based nanocomposites significantly reduced the ORAC and DPPH in roots and leaves under salt stress, compared to untreated plants. Biochar and biochar-based nanocomposite treatments significantly improved the DPPH activity in safflower roots and leaves under non-saline condition, but these treatments did not alter the ORAC. Application of biochar-based nanocomposites improved the DPPH activity under different levels of

Table 4 Reactive oxygen species, detoxification capacity and lipid peroxidation of safflower plants for the interaction of salinity and biochar-based nanocomposites

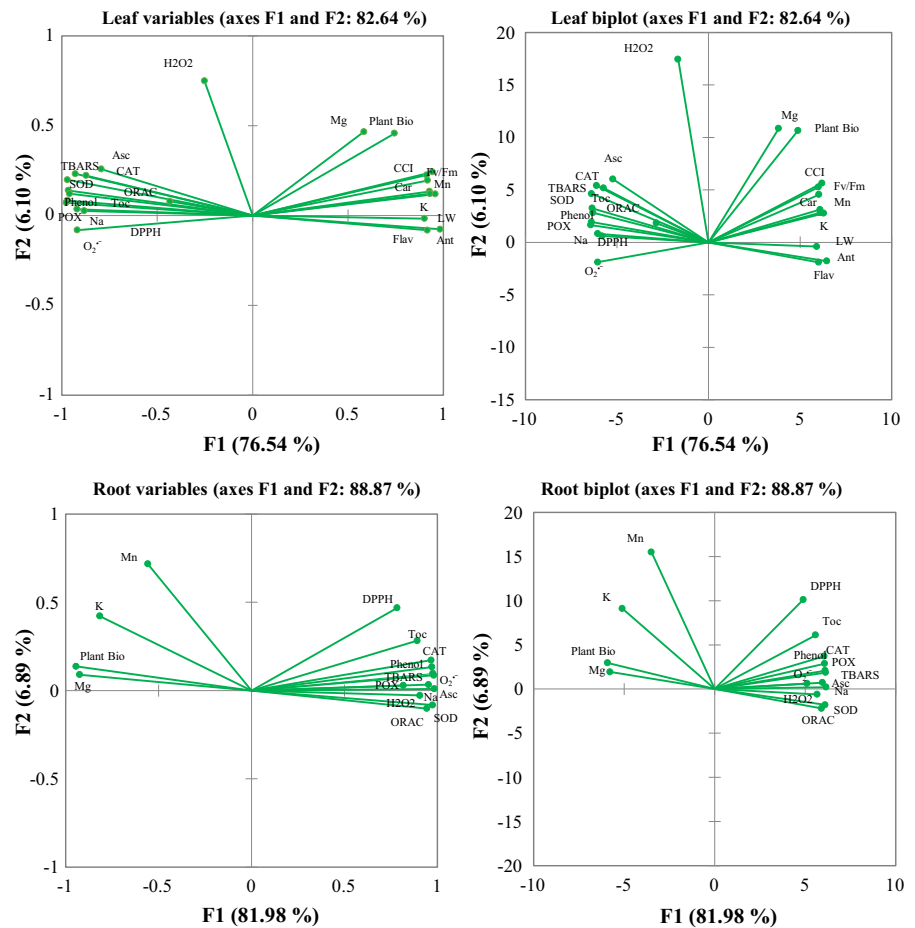
Salinity	Soil treatments	H_2O_2 ($\mu\text{mol g}^{-1}$ FW)		$O_2^{\bullet-}$ ($\mu\text{mol g}^{-1}$ FW h^{-1})		DPPH (%)		ORAC ($\mu\text{mol TE g}^{-1}$ DW)		TBARS (mmol g^{-1} FW)	
		Root	Leaf	Root	Leaf	Root	Leaf	Root	Leaf	Root	Leaf
0 (dSm $^{-1}$)	Non-biochar	0.12 $\pm$ 0.01 h	0.23 $\pm$ 0.04 h	0.14 $\pm$ 0.01 gh	0.18 $\pm$ 0.01 g	53.66 $\pm$ 0.67 j	62.33 $\pm$ 1.45 i	1890 $\pm$ 5.77 d	2158 $\pm$ 4.92 e	4.07 $\pm$ 0.08 h	4.04 $\pm$ 0.07 h
	Biochar	0.13 $\pm$ 0.02 h	0.17 $\pm$ 0.01 h	0.11 $\pm$ 0.02 h	0.21 $\pm$ 0.02 g	62.33 $\pm$ 0.88 i	67.33 $\pm$ 1.42 h	1876 $\pm$ 3.05 d	2233 $\pm$ 3.17 e	3.78 $\pm$ 0.12 h	3.88 $\pm$ 0.06 h
	BNC-MgO	0.13 $\pm$ 0.02 h	0.18 $\pm$ 0.01 h	0.13 $\pm$ 0.01 gh	0.18 $\pm$ 0.01 g	67.66 $\pm$ 0.89 h	68.66 $\pm$ 0.89 h	1865 $\pm$ 2.96 d	2179 $\pm$ 5.81 e	4.10 $\pm$ 0.03 h	4.03 $\pm$ 0.03 h
	BNC-MnO	0.12 $\pm$ 0.01 h	0.22 $\pm$ 0.04 h	0.14 $\pm$ 0.02 gh	0.18 $\pm$ 0.01 g	61.66 $\pm$ 0.85 i	67.00 $\pm$ 1.52 h	1890 $\pm$ 5.72 d	2186 $\pm$ 3.05 e	4.04 $\pm$ 0.03 h	4.22 $\pm$ 0.06 h
	BNC-MgO + BNC-MnO	0.12 $\pm$ 0.01 h	0.20 $\pm$ 0.02 h	0.13 $\pm$ 0.01 gh	0.21 $\pm$ 0.03 g	70.33 $\pm$ 0.34 gh	72.02 $\pm$ 1.47 g	1880 $\pm$ 5.81 d	2192 $\pm$ 19.40 e	3.88 $\pm$ 0.07 h	3.91 $\pm$ 0.16 h
6 (dSm $^{-1}$)	Non-biochar	0.74 $\pm$ 0.04 d	0.94 $\pm$ 0.03 d	0.38 $\pm$ 0.02 cd	0.46 $\pm$ 0.02 bc	90.31 $\pm$ 0.90 a	93.00 $\pm$ 1.59 b	3563 $\pm$ 31.79 c	4117 $\pm$ 9.21 c	7.44 $\pm$ 0.06 e	9.67 $\pm$ 0.17 e
	Biochar	0.51 $\pm$ 0.05 e	0.74 $\pm$ 0.03 f	0.30 $\pm$ 0.01 e	0.34 $\pm$ 0.01 ef	73.33 $\pm$ 0.87 fg	77.00 $\pm$ 1.44 f	3330 $\pm$ 15.38 c	3739 $\pm$ 20.08 cd	6.81 $\pm$ 0.13 f	7.62 $\pm$ 0.15 f
	BNC-MgO	0.34 $\pm$ 0.03 f	0.63 $\pm$ 0.03 g	0.26 $\pm$ 0.02 ef	0.32 $\pm$ 0.01 f	78.33 $\pm$ 1.20 d	82.33 $\pm$ 2.08 e	3231 $\pm$ 15.76 c	3543 $\pm$ 17.85 d	6.44 $\pm$ 0.07 g	7.11 $\pm$ 0.06 g
	BNC-MnO	0.36 $\pm$ 0.03 f	0.70 $\pm$ 0.03 f	0.25 $\pm$ 0.02 ef	0.30 $\pm$ 0.01 f	77.33 $\pm$ 0.33 de	83.00 $\pm$ 1.62 de	3324 $\pm$ 12.35 c	3611 $\pm$ 6.08 cd	6.80 $\pm$ 0.13 f	7.26 $\pm$ 0.16 fg
	BNC-MgO + BNC-MnO	0.26 $\pm$ 0.02 g	0.58 $\pm$ 0.02 g	0.20 $\pm$ 0.01 fg	0.29 $\pm$ 0.01 f	83.33 $\pm$ 0.35 b	87.00 $\pm$ 1.53 c	3131 $\pm$ 15.89 c	3500 $\pm$ 6.35 c	6.31 $\pm$ 0.12 g	6.82 $\pm$ 0.12 g
12 (dSm $^{-1}$)	Non-biochar	1.15 $\pm$ 0.02 a	1.31 $\pm$ 0.05 a	0.66 $\pm$ 0.05 a	0.72 $\pm$ 0.03 a	91.33 $\pm$ 0.32 a	95.00 $\pm$ 0.34 a	6162 $\pm$ 31.39 a	6859 $\pm$ 30.33 a	12.27 $\pm$ 0.15 a	16.69 $\pm$ 0.31 a
	Biochar	0.94 $\pm$ 0.03 b	1.14 $\pm$ 0.04 b	0.47 $\pm$ 0.05 b	0.48 $\pm$ 0.02 b	74.66 $\pm$ 1.45 ef	78.00 $\pm$ 1.55 f	5348 $\pm$ 28.91 b	6444 $\pm$ 29.41 ab	9.26 $\pm$ 0.16 b	14.59 $\pm$ 0.22 b
	BNC-MgO	0.86 $\pm$ 0.03 c	0.97 $\pm$ 0.03 cd	0.38 $\pm$ 0.04 cd	0.41 $\pm$ 0.02 cd	79.33 $\pm$ 1.20 cd	82.66 $\pm$ 1.45 de	5141 $\pm$ 30.23 b	6298 $\pm$ 50.85 b	8.58 $\pm$ 0.30 cd	13.60 $\pm$ 0.22 c
	BNC-MnO	0.84 $\pm$ 0.04 c	1.00 $\pm$ 0.02 c	0.41 $\pm$ 0.01 bc	0.39 $\pm$ 0.01 de	75.00 $\pm$ 1.52 ef	84.33 $\pm$ 2.34 d	5244 $\pm$ 29.35 b	6320 $\pm$ 15.16 b	8.81 $\pm$ 0.15 c	13.97 $\pm$ 0.24 c
	BNC-MgO + BNC-MnO	0.74 $\pm$ 0.03 d	0.84 $\pm$ 0.03 e	0.33 $\pm$ 0.01 de	0.32 $\pm$ 0.02 f	82.33 $\pm$ 0.65 bc	87.00 $\pm$ 1.00 c	5050 $\pm$ 28.86 b	6117 $\pm$ 16.18 b	8.31 $\pm$ 0.17 d	13.11 $\pm$ 0.07 d

Data represent the average of three replicates ($n = 3$) $\pm$ standard error.

H_2O_2 hydrogen peroxide; $O_2^{\bullet-}$ oxygen radicals; DPPH 2,2-diphenyl-1-picrylhydrazyl; ORAC oxygen radical absorbance capacity; TBARS thiobarbituric acid-reactive substances; TE trolox equivalent; FW fresh weight; BNC-MgO biochar-based nanocomposite of magnesium oxide; BNC-MnO biochar-based nanocomposite of manganese oxide

Different letters indicate significant differences by Duncan multiple range test at $p \leq 0.05$.

Fig. 3 Principal component analysis (PCA) of the leaf and root parameters of safflower plants. *Plant bio* plant biomass; *Na*:sodium; *K* potassium; *Mg* magnesium; *Mn* manganese; *Fv/Fm* efficiency of photosystem II; *RETR* relative photosynthetic electron transport rate; *CCI* chlorophyll content index; *Ant* anthocyanin; *Car* carotenoid; *Flav* flavonoid; *Toc* α -tocopherol; *SOD* superoxide dismutase; *CAT* catalase; *POX* peroxidase; *Asc* ascorbic acid; *TBARS* thiobarbituric acid-reactive substances; $O_2^{\bullet-}$ oxygen radicals; H_2O_2 hydrogen peroxide; *ORAC* oxygen radical absorbance capacity; *DPPH* 2,2-diphenyl-1-picrylhydrazyl; *LWC* leaf water content.



salinity in comparison with the solid biochar treatment. However, these nanocomposite treatments reduced the ORAC in saline conditions. In most cases, the highest value for DPPH was achieved by combined treatment of nanocomposites of magnesium and manganese.

Principal component analysis

PCA was used to evaluate the relation between different parameters of safflower plants. The PCAs were contributed up to 88% of variance in leaf and root parameters (Fig. 3). According to the PCA results, sodium content antioxidant enzymes, ascorbic acid and phenolic compounds, ROS generation and TBARS content (lipid peroxidation) were in the opposite direction, but potassium, magnesium, manganese, CCI, carotenoid, flavonoid, anthocyanin, Fv/Fm and RETR were in the positive direction with the

growth of safflower plants. Safflower biomass had a strong positive correlation with magnesium content, CCI, Fv/Fm and RETR values. Additionally, RETR value was highly correlated with CCI and carotenoid content. Carotenoid, leaf water and TBARS content were more effective than other parameters in changing safflower growth. TBARS content showed a strong correlation with phenolic compounds in root and CAT and SOD enzymes activities in leaf tissues. In ROS group, $O_2^{\bullet-}$ generation was more effective than H_2O_2 on lipid peroxidation in safflower roots and leaves. DPPH activity showed a strong correlation with α -tocopherol content and POX activity in safflower roots and leaves, respectively. ORAC was closely correlated with the SOD activity in both root and leaf tissues.

Discussion

The high level of ROS generation in safflower due to salt toxicity could be associated with the hindrance in electron transfer rate in thylakoid membrane and efficiency of photosystem II. While the chlorophylls are the chief pigments in the light harvesting and photosynthetic process in plants, other photopigments such as anthocyanin, flavonoid and carotenoid have also vital defensive roles in regulating electron transports and energy through the photosystems. These pigments have also a critical effect on photosystem assembly and energy gathering by harvesting light energy in an area of the visible range of light, where chlorophyll efficiency is lower as a result of diminishing the transport of energy to chlorophyll. Anthocyanin, flavonoid and carotenoid also protect the plants from extra light energy via energy downfall, membrane stability and ROS detoxification (Sakuraba et al. 2010). Decrements in photosynthetic pigments, relative electron transfer rate (RETR) and photosystem II efficiency (F_v/F_m) (Table 3) were steered to an increase in $O_2^{\bullet-}$ and H_2O_2 generation, non-enzymatic antioxidant compounds and antioxidant enzymes activities under salt toxicity of soil.

Biochar-based metal oxide nanocomposites had an excellent sodium sorption capacity and cation exchange capacity, which can reduce sodium uptake by plants and enhance potassium, magnesium and manganese contents in plant roots and leaves (Table 2). The increment of sodium sorption capacity of biochar in the composition process could be related to the enhancement of specific surface area and pore volume (Table 1). The specific surface area and pore volume of biochar are important factors in controlling sodium sorption capacity of biochar (Rostamian et al. 2015). The improvements of potassium, magnesium and manganese contents in plant tissues are the result of CEC enhancement and nutrient bioavailability in salt-stressed plants (Table 2). Accelerating nutrients absorption by biochar-based metal oxide nanocomposites are related to high specific surface area of these materials (Table 1).

The results of this study clearly showed that enhancing salt toxicity of soil decreased safflower growth by increasing ROS generation (Tables 3 and 4) and decreasing photosynthetic pigments such as chlorophylls, flavonoid, carotenoid and anthocyanin in safflower leaves (Table 3). The severe level of salt

toxicity in plant cells was also found to increase the ROS generation ($O_2^{\bullet-}$ and H_2O_2) in plant cells, consistent with other studies (Ali et al. 2017). The oxidative stress of salt toxicity degraded photosynthetic pigments, leading to a decreased growth of safflower (Table 3). Plant cells are typically protected from lipid peroxidation by antioxidant systems such as non-enzymatic compounds and enzymes which remove ROS. High level of $O_2^{\bullet-}$ and H_2O_2 in plant cells can cause a lethal effect on lipid peroxidation, protein and chlorophyll degradation and nucleotide mutation. Enhancing the antioxidant enzymes activity and non-enzymatic antioxidant compounds (Figs. 1 and 2) in response to salt toxicity is a common tool to detoxify ROS. Increasing activity of antioxidant enzymes, non-enzymatic compounds and ROS detoxification potential with enhancing salt toxicity is related to a high generation of ROS in safflower roots and leaves (Table 4). Salt toxicity can also reduce leaf water content and plant growth by reducing water uptake (Table 3).

Addition of biochar-based nanocomposites of magnesium and manganese oxides and just biochar to the soil reduced the ROS generation, lipid peroxidation and non-enzymatic antioxidant compounds and enzymatic antioxidant activities (Figs. 1 and 2; Table 4) through increasing the synthesis of photosynthetic pigments, relative photosynthetic electron transport rate and efficiency of photosystem II and decreasing sodium accumulation under salt stress (Table 2). Carotenoid and chlorophyll contents showed the highest correlation with relative photosynthetic electron transport rate and efficiency of photosystem II in leaf tissues (Fig. 3). Enhancing relative photosynthetic electron transport rate and maximum quantum yield of photosystem II by biochar-based nanocomposites of magnesium and manganese oxides was also related to the augmentation of carotenoid and chlorophyll biosynthesis in plant cells (Table 3; Fig. 3). Improving maximum quantum yield of photosystem II by biochar and biochar-based nanocomposites of magnesium and manganese oxides enhanced the photosynthetic electron transport rate under salinity (Table 3).

Biochar and biochar-based nanocomposites decreased the antioxidative activities in safflower roots and leaves by diminishing ROS generation under salt stress, since the antioxidative activities and ROS generation were closely related to each other (Fig. 3).

Biochar and biochar-based nanocomposites reduced sodium uptake in safflower plants (Table 2). This mechanism is related to the sodium absorption power of char materials (Table 1). Increasing enzymatic antioxidative activities and non-enzymatic antioxidant compounds by nanocomposites in comparison with biochar was probably associated with enriching plant cells by magnesium and manganese ions (Table 2). The most important function of magnesium and manganese is the activation of enzymes at different physiological pathways. Biochar-based nanocomposite of magnesium oxide resulted in maximum CCI. Magnesium is a component of chlorophyll molecule, and obviously application of this nutrient can increase the CCI of plants. Biochar-based nanocomposite of manganese also increased CCI, compared to the non-biochar and solid biochar treatments. Manganese is indirectly related to the formation of chlorophyll molecules and has some important roles in the evolution of oxygen in photosynthesis, oxidation–reduction reactions and electron transport systems in plant cells (Fageria 2016).

The lowest TBARS in plants treated with biochar-based nanocomposites of magnesium + manganese oxides under salt stress (Table 4) could be attributed to the highest content of phenolic compounds (compared with biochar and individual nanocomposite treatments) and the lowest generation of $O_2^{\bullet-}$ in both roots and leaves of safflower. Phenolic compounds and $O_2^{\bullet-}$ generation showed the high correlation with TBARS in this research (Fig. 3). Application of biochar-based nanocomposites of magnesium + manganese oxides had a better effect than other treatments on rising carotenoid content, CCI, maximum quantum yield of photosystem II, photosynthetic electron transport rate and DPPH activity in plant cells, leading to the highest plant biomass under saline and non-saline conditions (Table 3). Since carotenoid content, CCI, maximum quantum yield of photosystem II and photosynthetic electron transport rate in plant cells were positively correlated with plant biomass (Fig. 3), adding biochar-based nanocomposites of magnesium and manganese oxides to the soil was the most effective way for improving salt tolerance and growth of safflower under different levels of salinity.

Conclusions

Salt toxicity of soil reduced safflower growth by increasing sodium accumulation and diminishing maximum quantum yield of photosystem II, photosynthetic electron transport rate and biosynthesis of photosynthetic pigments. Application of biochar and particularly biochar-based nanocomposites of magnesium and manganese oxides reduced sodium accumulation and improved the quantum yield of photosystem II, photosynthetic electron transport rate and the biosynthesis of photosynthetic pigments, thereby reduced the ROS generation and oxidative stress in safflower plants under salt stress. These reductions of oxidative stress were related to the high leaf chlorophyll and carotenoid contents in safflower plants and better electron transfer rate under salt stress. The results of this research indicated that enriching biochar by different nutrients could be helpful in improving growth and stress tolerance of plants. Similar studies on different crops may further reveal the beneficial effects of biochar-based metal oxide nanocomposites on plant performance under different environmental conditions.

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Compliance with ethical standards

Conflict of Interest The authors declare that they have no conflict of interest.

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